IEEE SMC-B2S

Technical Committee on
Bio-mechatronics and Bio-robotics Systems

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MENU

IEEE Transactions on Cognitive and Developmental Systems

Special Issue on

Neuro–Robotics Systems： Sensing, Cognition, Learning and Control

AIM AND SCOPE

Neuro–robotics Systems (NRS) is a combined study of implementing human–like sensing, sensorimotor learning, coordination, cognition and control in autonomous robots, which can be integrated with cognitive capabilities, allowing them to imitate the way of humans and other living beings. NRS, a branch of neuroscience within robotics, is the current state–of–the–art research, as well as an important pillar in many countries' brain projects. It assists the next generation of robots with embodied intelligence to be aware of themselves, interact with the environment and behave harmoniously with/as human beings. Therefore, it is a study to integrate recent breakthroughs in brain neuroscience, robotics and artificial intelligence in terms of new principles of understanding, modeling and developing robotic systems. This way of implementation will introduce smart and straightforward configuration of autonomous robots capable of handling complex tasks and adapting to unstructured environments. It will enable robots or robotic devices to not only do much more work, but also be smart enough to support or augment human abilities. As a bridge between neuroscience and robotics, it encourages researchers to study and understand how to define and develop the "brain" for future robots.

THEMES

This special issue aims at surveying the state of the art of the latest breakthrough technologies, new research results and developments in the area of NRS. We are particularly interested in papers that describe the formulation of various functions of NRS, including human-like sensing, fusion, cognition, learning and control, especially, the topics related to system sensing, multi-dimensional information fusion, and cognitive computation, sensorimotor learning and control technology. It provides a platform for interdisciplinary researchers to present their findings and latest developments of biomimetic mechatronics and robotics systems, covering relevant advances in engineering, computing, arts and bionic sciences. Areas of interest include, but are not limited to

- Multi-modal perception, communication and interaction
- Multi-modal neuromorphic computing
- Brain-inspired end-to-end perception and control in robots
- Knowledge representation, information acquisition, and decision making in neuro-robotics systems
- Cognitive mechanism, and intention understanding in neuro-robotics systems
- Affective and cognitive sciences for bio-mechatronics
- Augmented cognitive robot systems, neuro-mechanical systems
- Biomimetic modeling of perception and control in neuro-robotics systems
- Brain-inspired development of rehabilitation robot systems, medical healthcare robot systems, prosthetic device systems, assistive robot systems, wearable robot systems for personal cooperative assistance.
- Sensorimotor coordination and control
- Multi-modal intelligent learning and skill transfer system for multiple neuro-robotic systems

- Robotic application oriented brain-inspired artificial intelligence algorithms and platforms on modeling, sensing, cognition, learning and control.

SUBMISSION

Manuscripts should be prepared according to the "Information for Authors" of the journal found at http://cis.ieee.org/publications.html and submissions should be done through the IEEE TCDS Manuscript center: https://mc.manuscriptcentral.com/tcds-ieee and please select the category "SI: **Neuro-Robotics Systems**".

IMPORTANT DATES

Deadline for manuscript submissions	30. November 2017
Notification of authors	28. February 2018
Deadline for revised manuscripts	31. May 2018
Final decisions	31. July 2018
Special Issue Publication	February/March 2019

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